

Kushal Madappa

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PROFILE

MSc Data & Computational Science student at University College Dublin, with a First Class Honours (1:1) B.E. in Computer Science and Business Systems. First author of a peer-reviewed Springer paper on deep learning for medical imaging, which received the conference Best Paper Award. Nine months' industry experience at LG Soft India as a research intern. Strong foundation in statistical modelling, machine learning and programming in Python and R.

Work authorisation: Stamp 2, eligible to work 20 hours per week during term and 40 hours per week during scheduled holiday periods. Eligible to apply for the Third Level Graduate Programme (Stamp 1G) on completion of the MSc.

Seeking: Part-time or internship data science roles during the MSc, and graduate data scientist positions from September 2027.

EDUCATION

MSc in Data & Computational Science 2026 – 2027

University College Dublin (UCD), Dublin, Ireland

Modules: Data Programming with Python, Data Programming using R, C Programming, AI & Machine Learning, Statistical Machine Learning, Exploratory Data Analytics, Foundations of Statistics, Cloud Computing.

B.E. in Computer Science and Business Systems 2021 – 2025

JSS Science and Technology University (JSSSTU), Mysuru, India

First Class Honours (1:1), CGPA 9.31/10, Top 5% of cohort; highest marks in Computational Statistics and in Cloud, Microservices & Applications

Relevant modules: Linear Algebra, Machine Learning, Artificial Intelligence, Design & Analysis of Algorithms, Data Structures, Internet of Things, Probability & Calculus, Business Strategy, Financial and Marketing Management

TECHNICAL SKILLS

- **Languages:** Python, R, SQL, C
- **ML & Statistics:** statistical analysis, scikit-learn, TensorFlow, Keras, pandas, NumPy, Matplotlib
- **Data & Backend:** MySQL, SQL Server, FastAPI, REST API integration
- **Cloud:** AWS (EC2, S3, RDS, IAM, VPC, ELB), Google Cloud Platform
- **Tools:** Git, GitHub, Docker, Jupyter Notebook, VS Code, Anaconda, Excel
- **LLMs & AI:** Anthropic and OpenAI APIs, LangChain, RAG pipelines, prompt engineering
- **Dashboards:** Power BI, Tableau

EXPERIENCE

Freelance Full-Stack Developer - PhotoBook Pro Mar 2026 – Jul 2026

Contract project for an independent retail client, Mysuru, one of three developers

An e-commerce platform where customers design custom photobooks in a browser-based editor, order them, and have them printed by a network of partner print labs and shipped with live courier tracking.

- Owned the delivery subsystem end to end (*Python, FastAPI, SQLAlchemy, PostgreSQL*): shipment data model, a client for the Shiprocket courier API (AWB assignment, label and invoice generation), and a webhook receiver with HMAC verification and idempotency guards that kept tracking correct under duplicate and out-of-order callbacks.
- Built the automatic album generation flow (*TypeScript, Next.js, Fabric.js*), assembling complete page layouts off-screen from uploaded photos before handing off to the editor.

V2X Engineer Intern - LG Soft India Pvt. Ltd., Bengaluru Jul 2024 – Mar 2025

Vehicle Component Solutions (VCS) division, V2X (Vehicle-to-Everything) team

- Built a Python automation tool that scans large log directories to locate specific entries and identify their source file; still in daily use by the test team and cut a previously manual search task to seconds.
- Validated V2X communication protocol implementations and debugged software modules across the automotive communication stack. Worked within an agile team of engineers over a nine-month placement, the longest internship offered in the division.

PROJECTS

Brain Tumour Classification from MRI *Python, TensorFlow, Keras, Flask*

- Built a two-stage pipeline on a 7,023-image MRI dataset: VGG19-UNet segmentation to isolate tumour regions, then EfficientNetB1 classification across four classes (glioma, meningioma, pituitary, no tumour).
- Baseline CNN reached 96% test accuracy; EfficientNetB1 with transfer learning reached near-perfect accuracy. Investigating that gap highlighted the risk of patient-level leakage in the public train/test split, an issue I would address with grouped cross-validation at patient level.
- Deployed the model behind a Flask interface allowing MRI upload and inference.
- Published at ICACECS 2025 (Springer) and awarded Best Paper.

Malware Detection Web Application *Python, scikit-learn, pandas, Flask*

- Built an end-to-end classifier to distinguish malicious from benign Windows executables using static file features extracted without running the binaries, avoiding the overhead and risk of dynamic sandbox analysis.
- Handled preprocessing across a large feature set: cleaning malformed entries, encoding categorical header fields, and addressing class imbalance so the model was not biased toward the majority class.
- Selected HistGradientBoosting for its native handling of missing values and strong performance on high-dimensional tabular data, benchmarking it against simpler baselines before settling on the final model.
- Evaluated with precision and recall rather than accuracy alone, treating false negatives as the costlier error, an undetected malicious file matters more than a benign file flagged for review.
- Deployed the trained model behind a Flask web interface accepting `.exe` uploads and returning a classification with confidence, making the model usable by non-technical users.

Gesture-Based Volume Control *Python, OpenCV, MediaPipe*

- Developed a real-time computer vision application controlling system audio through hand gestures captured on a standard webcam, removing the need for keyboard or hardware controls.
- Used MediaPipe hand-landmark detection to track finger positions frame by frame, mapping the distance between thumb and index finger to a continuous volume range.
- Implemented smoothing and thresholding on the raw landmark stream to suppress jitter and prevent unintended volume changes from brief detection noise.
- Optimised the processing loop to sustain responsive frame rates on consumer hardware, keeping input-to-response latency low enough to feel immediate.

PUBLICATION

Madappa, K., Rutvik Sharma, G., Keerthi Vasan, R., Ankith, L., Nagarjun, A. (2026). *A Hybrid Deep Learning and U-Net Architecture for Automated Brain Tumor Detection and Classification*. In: Proceedings of the Sixth International Conference on Advances in Computer Engineering and Communication Systems (ICACECS 2025). Smart Innovation, Systems and Technologies, vol. 489, pp. 503–512. Springer, Cham. doi.org/10.1007/978-3-032-22118-6_45

Awarded Best Paper at ICACECS 2025, VNR Vignana Jyothi Institute of Engineering and Technology, Hyderabad; presented by the first author.

CERTIFICATIONS & ACHIEVEMENTS

- Arcade Legend (highest tier), Google Cloud SkillBoost - Advanced cloud engineering and data science labs
- Google Data Analytics Professional Certificate - Google (*in progress*)
- Python for Data Science and Machine Learning Bootcamp - Jose Portilla
- Introduction to Cloud Computing on AWS - Neal Davis
- Google Cloud SkillBoost: Cloud Engineering Fundamentals - Google